

EMAIL THREAT FORENSIC INVESTIGATION REPORT

OFFICIAL DIGITAL FORENSIC EXAMINATION DOSSIER // LAW ENFORCEMENT & SOC RECORD

Case Identifier	Examination Date	Investigation Status	Evidence Hash (SHA-256)
CASE-007	2026-09-04	VERIFIED EXAMINATION	5a9d2e6b1c8f0472d5a8e3c0f6b2...
Classification: MARKETING	Risk Score: 18 / 100	Sample: Weekly_Product_Di...	File Size: 412 KB

1.0 EXECUTIVE SUMMARY

A forensic examination was conducted on evidence sample 'Weekly_Product_Digest_Issue_214.eml' cataloged under reference CASE-007. The electronic message claims transmission from sender 'LumenForge Product' (digest@mail.lumenforge.in). Automated and deterministic protocol evaluation assessed an authoritative risk score of 18 out of 100, resulting in a forensic threat classification of MARKETING. Authentication protocol verification recorded SPF: PASS, DKIM: PASS, and DMARC: PASS. No direct lookalike domain registration was detected in the immediate IOC telemetry. Routing telemetry analyzed 3 relay hops from origin to terminal destination server.

2.0 EMAIL & EVIDENCE SPECIFICATIONS

Evidence Field	Forensic Data Value	Verification Finding
Original Filename	Weekly_Product_Digest_Issue_214.eml	Binary structure parsed and pres...
File Size	412 KB	Standard envelope size
Cryptographic Hash	5a9d2e6b1c8f0472d5a8e3c0f6b2d9a4e1c7f3b0d8a5e2c...	SHA-256 cryptographic integrity ...
Claimed Sender Name	LumenForge Product	Header-extracted display name
Claimed Sender Email	digest@mail.lumenforge.in	Header From address
Terminal Mail Server	Hyderabad, India	Terminal destination host
Originating Node	198.2.185.44	First recorded external IP hop

3.0 AUTHENTICATION PROTOCOL ANALYSIS

Standard email authentication protocols were checked against published Domain Name System (DNS) records. Discrepancies indicate that the sending infrastructure was not authorized to transmit on behalf of the domain.

Protocol	Verdict	Technical Examination Finding
SPF (Sender Policy Framework)	PASS	Sending host verified against authorized SPF record
DKIM (DomainKeys Identified)	PASS	Valid cryptographic signature verified against DNS public key
DMARC (Domain Alignment)	PASS	Message satisfies alignment policy for domain

4.0 HEADER & ROUTING INFRASTRUCTURE

The message envelope traversed 3 recorded network transit hops. Network infrastructure was cross-referenced with autonomous system records and datacenter threat feeds.

Hop	IP Address	Geographic Location	Relay Timestamp	Forensic Infrastructure Note
#1	198.2.185.44	Atlanta, United States	2026-09-06 11:05:00	Mailchimp commercial ASN
#2	198.2.177.12	Chicago, United States	2026-09-06 11:05:08	Bulk ESP relay
#3	202.83.21.45	Hyderabad, India	2026-09-06 11:05:35	Destination mail server

5.0 LANGUAGE, INTENT & CONTENT ANALYSIS

Content linguistics and social engineering vectors were scored at 4 out of 30 maximum allocated points. Extracted textual phrases were analyzed for deception patterns, urgency cues, and financial solicitation. No explicit semantic NLP phrase extractions cataloged for this sample.

6.0 REPUTATION & DOMAIN INTELLIGENCE

Attribute	Observed Evidence	Analyst Interpretation
Domain Age	4382 days	Domain age within normal operational thresholds
Typosquatting Target	None observed	No typosquatting target observed
Reputation Score Contribution	6 / 15 pts	Reputation scores reflect standard operational...

7.0 INDICATORS OF COMPROMISE (IOCs)

The following actionable technical indicators were extracted during examination for perimeter blocklisting and SIEM correlation:

Indicator Type	Technical Value	Context / Threat Finding
IP Address (Origin)	198.2.185.44	Originating relay host observed in he...
Email Address	digest@mail.lumenforge.in	Header From address
File Hash (SHA-256)	5a9d2e6b1c8f0472d5a8e3c0f6b2d9a4e1c7f3b0d8a5e...	Unique cryptographic hash of message ...

8.0 ATTRIBUTION ASSESSMENT

ATTRIBUTION FINDING: NO ATTRIBUTION CONCERNS

No anomalous attribution signatures were identified in available telemetry.
Note: Attribution conclusions represent technical forensic assessments based on available telemetry and do not constitute definitive legal identification without formal judicial subpoena records.

9.0 CORRELATION & INFRASTRUCTURE RELATIONSHIPS

Graph analysis cross-referenced entities across historical telemetry clusters and previous incident dossiers:

Source Entity	Target Entity	Correlation Basis / Linkage
email1	ip1	mailchimp outbound
email2	ip1	same esp pool
domain1	ip1	authenticated bulk sender

10.0 RISK ASSESSMENT & SIGNAL WEIGHTING

Overall Risk Score: 18 / 100 (Threshold evaluation: LOW RISK). The score is derived across four standardized examination quadrants:

Quadrant Vector	Allocated Weight	Observed Score	Analytical Contribution
Header & Routing	30 pts	8 pts	Evaluates relay anomaly, ASN hosting, and h...
Authentication Failure	25 pts	0 pts	Evaluates SPF, DKIM signature validity, and...
Language & Intent	30 pts	4 pts	Evaluates urgency, financial diversion, and...
Reputation & History	15 pts	6 pts	Evaluates domain age, typosquatting targets...

11.0 KEY INVESTIGATION FINDINGS

- 1. Low-risk baseline: Overall threat telemetry satisfies benign operational profile.
- 2. Authentication verified: Cryptographic signatures and domain alignment passed checks.
- 3. Domain registration integrity: No deceptive lookalike typography detected.
- 4. Origin routing: Outbound transmission originated from standard commercial/workspace relay.

12.0 RECOMMENDED INVESTIGATION ACTIONS

- 1. Routine Monitoring: Retain evidence hash in SOC telemetry archive for longitudinal baseline monitoring.
- 2. No Blocking Required: Communication parameters match verified corporate or standard commercial baselines.

13.0 AUTOMATED NARRATIVE GENERATION ARCHITECTURE NOTE

SYSTEM NOTICE: This document was produced via deterministic evidence extraction from validated email forensic JSON. The platform architecture is designed to integrate text-generation language models in future backend iterations to draft extended contextual DFIR narratives while maintaining the underlying JSON as the single source of truth.

14.0 EVIDENCE ATTESTATION & LEGAL DISCLAIMER

REPORT IDENTIFIER: REP-CASE-007 | PLATFORM: Email Threat Intelligence & Forensic Platform | EVIDENCE SOURCE: Ingested MIME/MSG Artifact | STATUS: VERIFIED FORENSIC RECORD

LEGAL DISCLAIMER: This document is an analytical examination record compiled from electronic message headers, cryptographic authentication tokens, DNS telemetry, and network routing hops. Findings should be corroborated with original raw headers, provider server logs, and judicial legal process before presentation in legal proceedings.